

# BENJAMIN RIOT-BRETÊCHER

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## EDUCATION

**Ph.D - Atmospheric and Oceanic Sciences**, McGill University 2022 -

*Supervisor:* [Prof. Yi Huang](#)

Relevant Coursework: Atmospheric Dynamics, Atmospheric Radiation, Numerical Methods, Waves & Stability

ACHIEVEMENTS: Comprehensive Examination (February 2024); Proposal Examination (August 2025)

**B.S - Atmospheric and Oceanic Sciences**, Université du Québec à Montréal 2019 - 2022

## PROFESSIONAL EXPERIENCE

**Teaching Assistant** 2023 -

McGill University

- [Introduction to Atmospheric Sciences - ATOC181](#) (Fall 2023, Fall 2025)
- [Science of Storms - ATOC184](#) (Winter 2024, Winter 2026)

**Visitor Student** Oct 2024 - Jan 2025

[School of Atmospheric Sciences, Nanjing University](#)

**Treasurer** Aug 2023 - Aug 2024

Council of Atmospheric and Oceanic Sciences (CAOS)

**Field Campaign Involvement** 2022 - Present

McGill University

- [ECALOT Field Campaign](#) (Fall 2024)
- [WHAFFERS Field Campaign](#) (Winter 2025)
- [GHG Montreal Project](#) (Ongoing)
- [PONEX](#) (Winter 2026)

**Research Assistant** May 2022 - Aug 2022

McGill University

*Supervisor:* [Prof. Yi Huang](#)

**Research Assistant** June 2021 - Aug 2021

Université du Québec à Montréal

*Supervisor:* [Prof. Jean-Pierre Blanchet](#)

## PUBLICATIONS

- **B. Riot-Bretêcher**, Y. Huang, L. Liu, J. Gero, [Strengthening Atmospheric Greenhouse Effect at the Arctic North Slope of Alaska Site Evidenced by Long-Term Records of the Downwelling Longwave Radiance Spectrum](#), Journal of Geophysical Research: Atmospheres, 2025.

## CONFERENCES AND TALKS

- [Joint FORUM-HAWC Workshop](#) - **Poster** - Montreal, QC, Canada - October 2025
- [Gordon Research Conference on Radiation and Climate](#) - **Invited Speaker** - Lewiston, ME, USA - July 2025
- [Gordon Research Seminar on Radiation and Climate](#) - **Speaker** - Lewiston, ME, USA - July 2025
- [2nd ESA-JAXA EarthCARE In-Orbit Validation Workshop](#) - **Co-author, Presented by L. Liu** - Frascati, Italy - March 2025 Validation of EarthCARE cloud retrievals using surface spectral infrared radiances from the ECALOT campaign. (Liu, L., **Riot-Bretêcher**, B. et al.)
- [Nanjing University - Atmospheric Sciences Seminar](#) - **Speaker** - Nanjing, China - December 2024
- [43rd Canadian Symposium on Remote Sensing](#) - **Attendee** - Québec City, QC, Canada - July 2022
- [Canadian Meteorological and Oceanographic Society](#) - **Attendee** - Remote - June 2021

## AWARDS AND SCHOLARSHIP

- McGill Graduate Excellence Fellowship (2022 - 2028)
- Graduate Mobility Award - McGill University (Oct 2024 - Jan 2025) - \$ 3,800
- Canadian Meteorological and Oceanographic Society Undergraduate Scholarship (April 2021) - \$ 1,000

SKILLS

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| Technical Skills    | MATLAB, Fortran, Python, Java, R, Bash, Linux, Radiative Transfer Model, Weather Balloon Launches, AERI Maintenance, Optimal Estimation Method                                                                                                                                                                                                                                                                                                                                                                                                             |
| Mathematical Skills | Algebra, Calculus, Probability, Statistics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Certifications      | <ul style="list-style-type: none"><li>• Data Security - Compute Ontario (June 2025)</li><li>• Introduction to Advanced Research Computing - Compute Ontario (June 2025)</li><li>• Introduction to Version Control Using Git - Compute Ontario (June 2025)</li><li>• Scaling Up HPC Workflows - Compute Ontario (June 2025)</li><li>• Career Essentials in Software Development by Microsoft and LinkedIn (<a href="#">Certificate ID</a>)</li><li>• Programming Foundation: Beyond the Fundamentals by LinkedIn (<a href="#">Certificate ID</a>)</li></ul> |